[1] Two Moroccan women scientists appeared on the list of the AD (Alper-Doger) Scientific Index in 2023. It lists scientists and researchers internationally, based on their performance and scientific efforts. Rajaâ Cherkaoui El Moursli and Farida El Fassi, professors at Mohammed V University in Rabat, managed to appear on the AD index list for their great contributions to scientific research.

[2] El Moursli was born in Salé in 1954. As a student, she was fascinated and inspired by the American astronaut Neil Armstrong who became the first person to walk on the Moon in 1969. After getting her Bac in mathematics in Rabat, she went to study in Grenoble in France where she obtained her doctorate in physics. In 1982, she returned to Morocco and joined Mohammed V University as a teacher and researcher in nuclear physics, medical physics and high energy physics. From 2013 to 2017, she was the vice president of **the university**. In 2015, El Moursli received a UNESCO award for "Women in Science" for her key contribution to one of the discoveries in physics. In 2019, she was elected vice president of the Network of African Science Academies (NASAC), which aims to make science as a way for positive change in Africa.

[3] El Moursli was happy with her position among international scientists. It is an achievement, which awards a long journey of research, work and innovation. She considered her ranking an honor for Morocco and the efforts of researchers and their competencies. "It is a kind of scientific diplomacy that ensures the popularity of Morocco in international academic forums," she explains.

[4] Farida El Fassi, a professor of nuclear physics at Mohammed V University in Rabat, ranked second in the Arab world and in Africa on the AD Index. When she was a child, she had a strong passion for sciences. "When I was little, my parents nurtured my curiosity and problem-solving skills through practicing home tasks such as cooking, doing household chores and planning trips," she explains. El Fassi was born in Larache where she attended middle and high school before moving to Tetouan to complete her university studies. After that, she joined the University of Valencia in Spain and got a doctorate in particle physics **there**. She then worked as a researcher at the National Center for Scientific Research in Spain. In 2014, she returned to Morocco with the aim of spreading scientific knowledge among Moroccan students.

[5] El Fassi believes that scientific research "requires individual effort as well as the support of the state. Therefore, Morocco needs to make more efforts to provide more research opportunities and suitable working conditions to minimize brain drain."

Adapted from: www.pledgetimes.com and www.twas.org

I. COMPREHENSION (15 POINTS)

BASE ALL YOUR ANSWERS ON THE TEXT.

A. COMPLETE THE CHART WITH INFORMATION FROM THE TEXT. (2 pts)

	Birthplace	Date of return to Morocco
Rajaâ Cherkaoui El Moursli	.....	.....
Farida El Fassi	.....	.....

B. ARE THESE STATEMENTS TRUE OR FALSE? JUSTIFY.

1. When El Moursli returned to Morocco, she started working as vice president of the university.

…….

2. El Fassi worked abroad before returning to Morocco.

…….

C. ANSWER THE FOLLOWING QUESTIONS. (3 pts)

1. Who was El Moursli's role model when she was young?

…….

2.Why did El Fassi decide to return to Morocco?

…….

3. What does El Fassi suggest to fight brain drain?

…….

D. COMPLETE THE SENTENCES WITH INFORMATION FROM THE TEXT. (2 pts)

1. The goal of the Network of African Science Academies is

…….

2. Doing household chores and planning trips allowed El Fassi to develop

…….

E. PICK OUT FROM THE TEXT PHRASES OR SENTENCES WHICH SHOW THAT (2 pts)

1. El Moursli worked hard for her position. (paragraph 3)

…….

2. EL Fassi loved sciences at an early age. (paragraph 4)

…….

F. FIND IN THE TEXT WORDS THAT MEAN ALMOST THE SAME AS (2 pts)

1. important (paragraph 2): . . . . . .

2. reduce (paragraph 5): . . . . . .

G. WHAT DO THE UNDERLINED WORDS IN THE TEXT REFER TO? (2 pts)

1. the university (paragraph 2): . . . . . .

2. there (paragraph 4): . . . . . .

A. GIVE THE CORRECT FORM OF THE WORDS IN BRACKETS. (2 pts)

1. Climate change needs a lot of (collaborate) . . . . . to reduce its negative effects.

2. Mourad is a very (competence) . . . . . manager. His leadership qualities are motivating.

B. FILL IN THE BLANKS WITH THE CORRECT WORDS FROM THE LISTS. (3 pts)

1. I can't help . . . . . whenever I watch Mr. Bean's shows.

laugh - laughing - to laugh

2. Khalid will have . . . . . the university in two years' time.

leave - leaving - left

3. The electric cars . . . . . Morocco will produce are environment friendly.

where - which - whose

C. FILL IN THE BLANKS WITH THE CORRECT WORDS FROM THE LIST. (2 pts)

get - keep on performing - keep up - training

1. Many jobs will disappear in the future as robots will replace humans in . . . . . some tasks.

2. Workers need more training programs to . . . . . with the rapid changes in the job market.

D. REWRITE THE SENTENCES AS INDICATED. (4 pts)

1. "I think you should convert the document to PDF?" she said.

She advised me . . . . . .

2. She exercised a lot. She failed to lose weight.

Despite . . . . . .

3. I didn't attend the online lesson because of poor internet connection.

I wish. . . . . .

4. The school created a new blog to share activities.

A new blog . . . . . .

E. CHOOSE THE MOST APPROPRIATE RESPONSE. (2 pts)

1. Ahmed: I'm afraid to say it, but I've been waiting for a long time to get my phone repaired!
Technician: . . . . . .

a. Definitely. b. I'm awfully sorry. c. It's my pleasure.

2. Your friend: Our Moroccan national football team has made it to the World Cup semifinals.
You: . . . . . .

a. Well-deserved! b. Poor you! c. What a pity!

F. COMPLETE THE FOLLOWING EXCHANGES APPROPRIATELY. (2 pts)

1. Nawal: I find this film boring.

You: I have the same feeling. (Suggest something else) . . . . . .

. . . . . .

2. Kamal: (Asks his friend to take a

picture) . . . . . .

. . . . . .

Friend: Of course.

DO TASKS 1 AND 2

TASK 1: Complete the following paragraph (4 pts)

Using surveillance cameras at school has many benefits. First, ...

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

TASK 2: (6 pts)

Nowadays, people think that living in big cities is very challenging.

Write an article to be published in your school e-magazine about the problems people face in big cities.

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

KEY AND MARKING SCALE

Accept any appropriate answer not mentioned in this key.

I. COMPREHENSION (15 POINTS)

Testees should not be penalised for spelling or grammar mistakes in the comprehension section.

CHART FILLING: (2 pts)

El Moursli : born in Salé; returned to Morocco in 1982.
El Fassi : born in Larache; returned to Morocco in 2014.

TRUE / FALSE: (2 pts)

1. **False.** "In 1982, she returned to Morocco and joined Mohammed V University as a teacher and researcher." (paragraph 2) She became vice president later (2013-2017).
2. **True.** "She then worked as a researcher at the National Center for Scientific Research in Spain." (paragraph 4)

WH QUESTIONS: (3 pts)

1. The American astronaut Neil Armstrong.
2. To spread scientific knowledge among Moroccan students.
3. She suggests that Morocco should make more efforts to provide more research opportunities and suitable working conditions.

SENTENCE COMPLETION: (2 pts)

1. ...to make science a way for positive change in Africa.
2. ...her curiosity and problem-solving skills.

PICK OUT: (2 pts)

1. "It is an achievement, which awards a long journey of research, work and innovation."
2. "When she was a child, she had a strong passion for sciences."

RIGHT WORD MEANING: (2 pts) 1. key 2. minimize

WORD REFERENCE: (2 pts) 1. Mohammed V University 2. (at) the University of Valencia (in Spain)

II. LANGUAGE (15 POINTS)

WORD FORMATION: (2 pts) collaboration · competent

GAP FILLING: (3 pts) 1. laughing 2. left 3. which

GAP FILLING: (2 pts) performing · keep up

REWRITING SENTENCES: (4 pts)

1. She advised me **to convert the document to PDF**.
2. Despite **exercising a lot, she failed to lose weight**.
3. I wish **I had attended the online lesson**.
4. A new blog **was created by the school to share activities**.

APPROPRIATE RESPONSE: (2 pts) 1. b. I'm awfully sorry. 2. a. Well-deserved!

EXCHANGES: (2 pts)

1. Let's watch a comedy instead. / Why don't we go out for a walk?
2. Could you take a picture of me, please?

III. WRITING (10 POINTS; task 1: 4 pts; task 2: 6 pts)

The following criteria should be respected while scoring the writing tasks.

Scoring criteria	Task 1	Task 2
Relevance of content to the task	2 pts	3 pts
Appropriateness and variety of vocabulary	1 pt	2 pts
Grammar and mechanics	1 pt	1 pt

Model answers (any appropriate answer is accepted)

Task 1: Using surveillance cameras at school has many benefits. First, cameras make students and teachers feel safe because they prevent fights and thefts. Second, they help the administration to find out quickly what happened when there is a problem. Moreover, students behave better when they know that someone can see them. Finally, cameras protect the school equipment from damage. For all these reasons, I believe that surveillance cameras are very useful at school.

Task 2: The Challenges of Living in Big Cities Nowadays, more and more people move to big cities to find jobs and better schools. However, life in these cities is not easy. First of all, traffic is a serious problem. Streets are crowded, so people waste hours every day to go to work or school. Moreover, pollution makes the air dirty and causes health problems such as asthma. Another challenge is the high cost of living: rents and prices are very high, and many families struggle to pay their bills. In addition, big cities are noisy and stressful, and people often feel lonely because they do not know their neighbors. In conclusion, big cities offer many opportunities, but they also bring many problems. Governments should build more public transport and green spaces to improve people's lives.